

Mechanics of Machines

Project Report

Title: Design and Fabrication of a Scotch Yoke Mechanism for Rotary-to-Reciprocating Motion Conversion



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Title

Design and Fabrication of a Scotch Yoke Mechanism for Rotary-to-Reciprocating Motion Conversion.

Abstract:

The Scotch Yoke Mechanism is a mechanical arrangement used to convert rotary motion into reciprocating linear motion or vice versa. The mechanism consists of a rotating crank pin mounted on a wheel that slides inside a slot in the yoke. As the wheel rotates, the crank pin moves along a circular path and forces the yoke to move back and forth in a straight line. The purpose of this project was to design and fabricate a working model of a Scotch Yoke Mechanism using cardboard materials. The model demonstrates the fundamental principles of motion conversion used in various industrial and engineering applications. Experimental observation confirmed smooth conversion of rotational motion into linear reciprocating motion, validating the theoretical operation of the mechanism.

Introduction:

Mechanisms are combinations of machine elements designed to transmit and transform motion. Among the various motion conversion mechanisms, the Scotch Yoke Mechanism is widely used where simple and efficient conversion between rotary and reciprocating motion is required.

Unlike a slider-crank mechanism, the Scotch Yoke directly converts motion through a sliding pin arrangement, resulting in fewer moving parts and a compact design. This project demonstrates the working principle of the Scotch Yoke through a low-cost cardboard model.

Objectives:

- To understand the working principle of the Scotch Yoke Mechanism.
- To study the conversion of rotary motion into reciprocating motion.
- To fabricate a functional mechanism using cardboard materials.
- To observe displacement characteristics of the reciprocating member.
- To relate theoretical concepts with practical implementation.

Apparatus used:

- Cardboard sheets
- Wooden pin (or wooden skewer)
- Adhesive glue
- Cutter/utility knife
- Ruler and measuring tape

- Pencil and marker
- Cardboard circular disc
- Cardboard guide frame
- Base platform

Construction:

- A rectangular cardboard base measuring approximately 30 cm × 15 cm was prepared to provide a stable foundation for the model.
- Two cardboard support frames were fabricated and fixed vertically on the base. These supports were designed to hold the rotating shaft and crank wheel securely.
- A circular disc was cut from cardboard and mounted on a central axis. This disc acted as the rotating crank wheel.
- A **wooden pin** was fixed eccentrically near the outer edge of the cardboard disc. The wooden pin served as the crank pin and was responsible for transmitting rotary motion to the yoke.
- A slotted yoke was fabricated from cardboard by cutting a straight elongated slot at its center. The dimensions of the slot were chosen to allow smooth sliding of the wooden pin.
- A guide channel was constructed from cardboard strips to restrict the yoke's motion to a straight reciprocating path.
- The wooden pin attached to the rotating disc was inserted into the slot of the yoke. Proper alignment was ensured so that the pin could move freely inside the slot without jamming.
- All parts were assembled using glue and reinforced with additional cardboard pieces where necessary to improve rigidity and durability.
- The disc was manually rotated, causing the wooden pin to travel in a circular path while sliding inside the slot, thereby producing reciprocating motion of the yoke.

Procedure:

- Place the Scotch Yoke model on a flat and stable surface.
- Inspect all components and ensure that the crank wheel, shaft, crank pin, and yoke are properly connected and aligned.
- Rotate the crank wheel slowly by hand in the clockwise direction.
- Observe the movement of the crank pin as it follows a circular path.
- Note that the crank pin slides within the slot of the yoke and forces the yoke to move linearly along its guide.

- Continue rotating the crank through one complete revolution (360°).
- Observe that one complete revolution of the crank produces one complete reciprocating cycle of the yoke.
- Measure the maximum forward and backward displacement of the yoke from its central position.
- Record the position of the yoke at different crank angles such as 0° , 90° , 180° , 270° , and 360° .
- Compare the observed motion with the theoretical motion of the Scotch Yoke Mechanism.
- Repeat the experiment several times to verify consistent operation and minimize observational errors.
- Analyze the results and confirm that rotary motion has been successfully converted into reciprocating linear motion.

Working Observation

Crank Angle	Position of Yoke
0°	Central Position
90°	Maximum Forward Displacement
180°	Central Position
270°	Maximum Backward Displacement
360°	Returns to Central Position

Results and Calculations:

Crank Angle ($^\circ$)	Yoke Displacement (mm)	Time (sec)	Velocity (mm/s)	Acceleration (mm/s)
0	0	0	2.25	0
45	2.25	1	2.25	-4.5
90	4.5	2	-2.25	0
135	2.25	3	-2.25	0
180	0	4	-2.25	0
225	-2.25	5	-2.25	4.5
270	-4.5	6	2.25	-2.25
315	-2.25	7	0	
360	0	8		

Graphs:

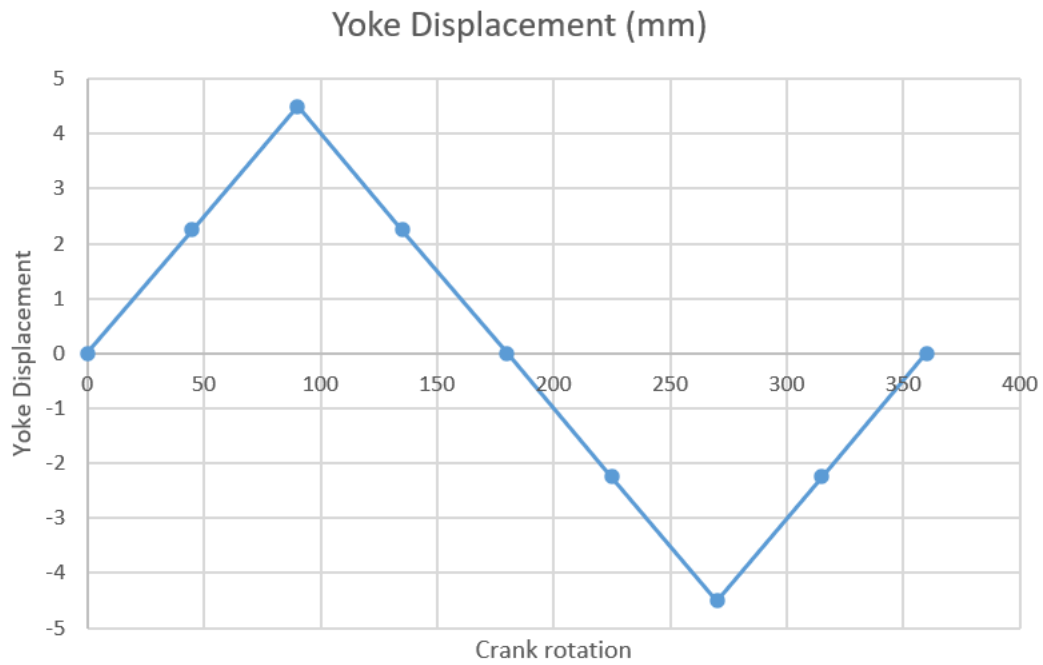


Figure 1

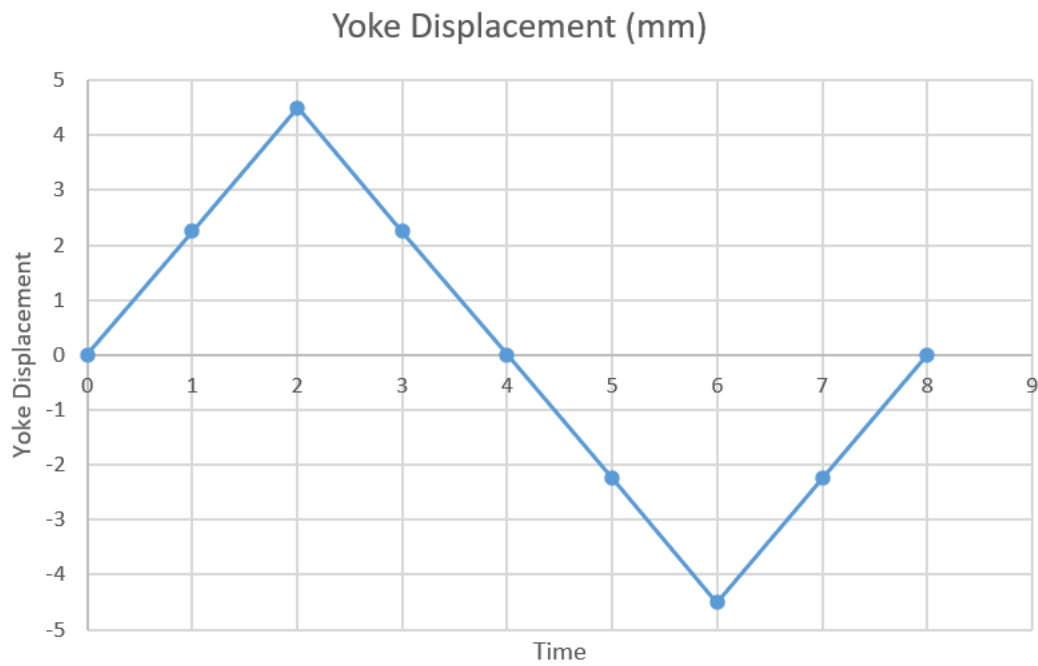


Figure 2

Velocity Graph

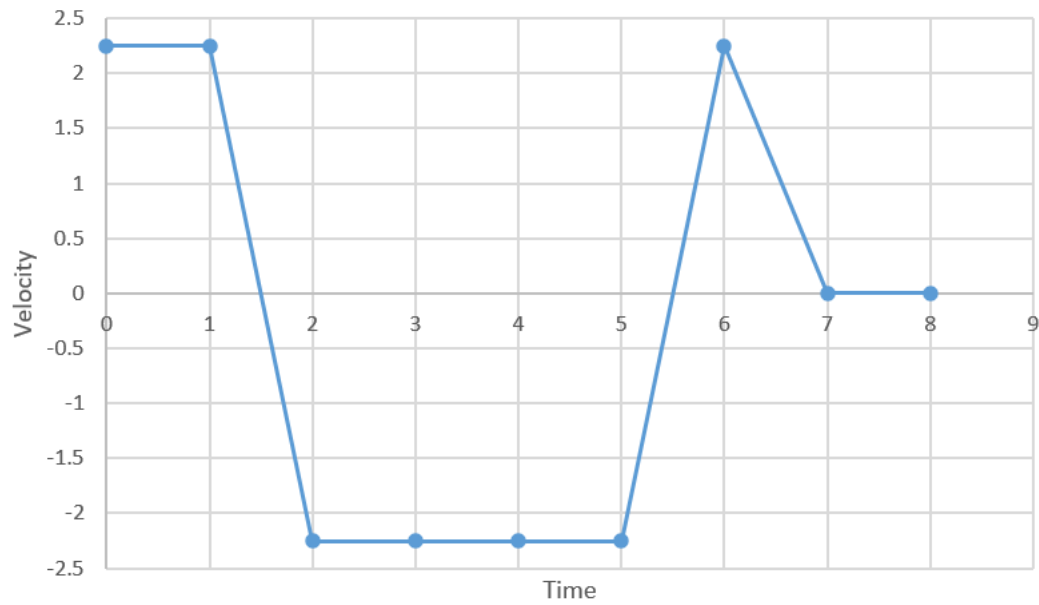


Figure 3

Acceleration Graph

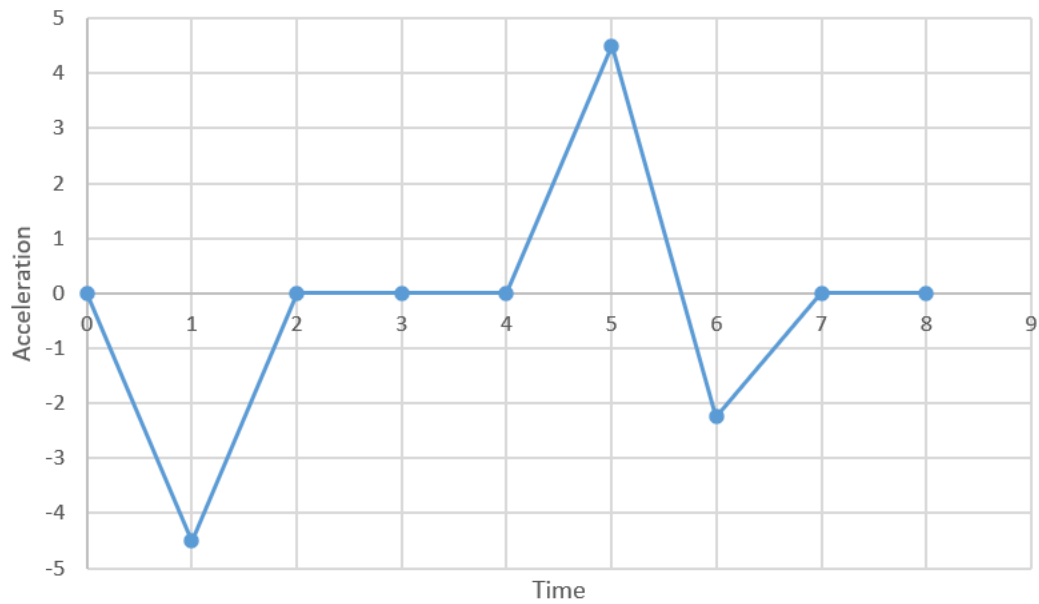


Figure 4

Discussion:

The Scotch Yoke model successfully demonstrated the conversion of rotary motion into reciprocating motion. The reciprocating slider moved smoothly within the guide as the crank rotated. The displacement pattern closely followed simple harmonic motion. Minor friction between cardboard surfaces slightly affected smoothness but did not alter the operating principle. The experiment verified the theoretical relationship between crank rotation and slider displacement.

Sources of Error

- Friction between cardboard surfaces.
- Misalignment of shaft and guide.
- Inaccurate dimensions during cutting.
- Loose joints and connections.
- Human error during measurements.
- Deformation of cardboard under repeated operation.

Precautions

- Ensure proper alignment of all components.
- Cut cardboard accurately according to dimensions.
- Secure all joints firmly.
- Avoid excessive force during operation.
- Maintain smooth surfaces to reduce friction.
- Verify free movement before testing.

Applications

The Scotch yoke mechanism is commonly used in:

- Control valve actuators in oil and gas pipelines.
- Reciprocating pumps.
- Shaping and slotting machines.
- Steam engines and hot-air engines.
- Internal combustion engine prototypes.
- Motion conversion systems in industrial automation.

Advantages:

- Simple design and construction.
- Fewer moving parts compared to slider-crank mechanisms.
- Compact structure.
- Produces exact sinusoidal displacement.
- Efficient conversion of rotary motion into reciprocating motion.

Disadvantages:

- High sliding friction between the pin and slot.
- Greater wear compared to connecting-rod mechanisms.
- Requires lubrication in practical applications.
- Not suitable for very high-speed continuous operation due to slot wear.

Limitations:

- High wear between pin and slot.
- Increased friction at high speeds.
- Not suitable for heavy loads.
- Requires proper lubrication in industrial applications.

Recommendations / Improvements

- Replace cardboard with acrylic or plywood for durability.
- Use ball bearings to reduce friction.
- Improve dimensional accuracy using CAD design.
- Add a motor drive for automatic operation.
- Install measurement scales for displacement analysis.
- Use stronger shafts and joints.

References

- https://en.wikipedia.org/wiki/Scotch_yoke
- <https://www.firgelliauto.com/blogs/mechanisms/scotch-yoke?srltid=AfmBOoq5guhMtEu3lonMBemgJVeHNRfPtyyX2uI-lho2AnhX04TMfh16>

- <https://testbook.com/mechanical-engineering/scotch-yoke-mechanism-application>

Conclusion

The Scotch Yoke Mechanism was successfully designed and fabricated using cardboard materials. The model effectively demonstrated the conversion of rotary motion into reciprocating linear motion through the interaction of a rotating crank pin and slotted yoke. The observed motion closely matched theoretical predictions and verified the principles of simple harmonic motion. The project enhanced understanding of machine mechanisms and provided practical experience in designing and assembling a working mechanical model. The objectives of the experiment were achieved successfully.